

Shah Imran Ahsan Chowdhury

GIS Analyst (PS), Esri Bangladesh, Level 16, Tower of Aakash, 54 Gulshan Avenue, Dhaka 1212, Bangladesh
imran.chowdhury2804@gmail.com, c.imran@esri.com

Research Interests

GIS, Remote Sensing (UAV, Satellite), GeoAI, Urban Informatics, Environmental Monitoring and Sustainable Urban Systems.

Education

Master of Science in Computer Science

Jahangirnagar University

April 2025 - Present

Dhaka, Bangladesh

Relevant Coursework & Labs: Digital Image Processing, Machine Learning, Deep Learning, Computer Vision, Programming Languages, Spatial Database.

Bachelor of Urban & Regional Planning

Rajshahi University of Engineering & Technology

January 2016 – February 2021

Rajshahi, Bangladesh

Relevant Coursework & Labs: Surveying and Cartography, GIS and Remote Sensing, Programming Techniques, Statistics for Planners, Computer Applications in Planning, Environmental Planning and Management, Urban Planning Studio.

Professional Experience

Esri Bangladesh

GIS Analyst (Professional Service Division)

July 2024- Present

Responsibilities: Closely working with Esri Malaysia, Esri Singapore, Esri Indonesia and Esri Australia to develop geospatial products.

Center for Environmental and Geographic Information Services (CEGIS)

Research Consultant (Remote Sensing Division)

March 2021- June 2024

Responsibilities: Satellite image processing. Creation of geoprocessing tool using Python, Python automation, Development of WebGIS dashboard as frontend developer, Geodatabase preparation, Hydrological analysis, LULC preparation, Spatial-Temporal Analysis, Conducting drone flight plan, UAV survey and analysis.

Academic Project

Prediction of Land Cover Change Pattern and Economic Valuation of

March 2020 - January 2021

Environmental Change: The Case of Chalan Beel, Bangladesh.

Nature of Work: Undergraduate Thesis, Department of Urban and Regional Planning, Rajshahi University of Engineering & Technology (RUET).

Research Activities: LULC map creation of dry and rainy season in 1999 and 2019, Land cover prediction through cellular automata (CA) and artificial neural network (ANN) algorithm, Land use pattern change observation, LULC matrix creation, Satellite based air quality measurement, Ecological service value prediction.

Outcome: Produced predictive LULC maps and estimated the future ecological service value of Chalan Beel, including community willingness-to-pay (WTP) analysis for its conservation and sustainable management.

Urban sprawl management through land readjustment scheme.

February 2018 – July 2018

Nature of Work: Site and Area Planning Studio Project, Department of Urban and Regional Planning, RUET.

Research Activities: Developed a comprehensive site plan layout to address urban sprawl using AutoCAD Map 3D, ArcGIS, SketchUp, and Lumion 3D. The study emphasized land readjustment principles.

Outcome: Produced a GIS-based spatial plan and 3D visual simulation video using Lumion.

Storm water management through effective landscaping: A case study on Rajshahi Medical College.

July 2017 - December 2017

Nature of Work: Landscape Planning Studio Project, Department of Urban and Regional Planning, RUET.

Research Activities: Designed a sustainable and resilient campus drainage and landscape system for Rajshahi Medical College by integrating GIS-based hydrological modeling with 3D visualization tools such as AutoCAD, SketchUp, and Lumion. The project emphasized eco-friendly spatial planning and data-driven design optimization to enhance both environmental performance and aesthetic harmony of the campus.

Outcome: Developed a fully 3D-modeled campus drainage and landscape design demonstrating functional stormwater management integrated with aesthetic and ecological design principles.

Professional Research Project

Next-Gen Transportation Surveying: YOLOv8 CNN Model and DeepSORT Tracking Algorithm for Vehicle Detection and Counting from UAV Footage.

October 2023 - November 2023

Nature of Work: Applied Research Project conducted at the Center for Environmental and Geographic Information Services (CEGIS) as part of a municipal masterplan initiative.

Research Activities: Developed AI based vehicle detection and tracking system using the YOLOv8 deep learning model integrated with the DeepSORT algorithm to process UAV footage. The project focused on automated traffic monitoring and enabling accurate vehicle counting. Achieved high detection precision through deep learning model and Python-based video analytics.

Outcome: Developed a functional system that automatically detects and counts vehicles in both directions of a highway and exports the results into an Excel-based summary report.

Ongoing Research

Hybrid GeoAI framework for post-classification building footprint validation using image and tabular feature fusion.

Nature of Work: Proposed MSc Research Project for Department of Computer Science and Engineering, Jahangirnagar University, Bangladesh.

Research Focus: Proposed development of a hybrid GeoAI framework that integrates two deep learning models: one for detecting building footprints from high-resolution imagery, and another for identifying and removing misclassified footprints using tabular feature-based classification. The study aims to enhance post-classification accuracy by fusing spatial and non-spatial features for robust building footprint validation.

Expected Outcome: Development of a hybrid GeoAI framework for accurate building footprint validation, contributing to MSc research project and future publication.

Post-Model Ensemble of UNet and DeepLabV3 Outputs for Enhanced Road Network Extraction: Mapping SDG Target 11.2 with UAV Imagery.

Nature of Work: Independent Research Project (Methodology under development).

Research Focus: Training multiple U-Net and DeepLabV3 models with different backbone architectures and ensembling their outputs within a GIS environment to evaluate accuracy enhancement in UAV-based road network extraction.

Expected Outcome: Preparation of a full-length research manuscript for submission to an international GeoAI or remote sensing journal upon completion of model experimentation.

Publications

Rashid, K.J., Tuli, R.D., Nasher, N.M.R., Akter, T., Karim, K.H.R., Hasan, M.M., Talha, M., **Chowdhury, S.I.A.** and Musharrat, M., 2024. *Greening and browning trend with physio-climatic drivers in Chattogram division, Bangladesh. Environment, Development and Sustainability*, pp.1–23. Springer Netherlands.

Biswas, R.N., Rashid, K.J., Ullah, M.A. and **Chowdhury, S.I.A.**, 2024. *Implications of Jhau (Tamarix: Casuarina equisetifolia) forest deforestation on coastal landscape ecology and climate change adaptation in Cox's Bazar, Bangladesh. Ecological Frontiers*, 44(5), pp.1027–1051. Elsevier.

Biswas, R.N., Islam, M.S., Miah, S., Lasker, M.S., Kar, S.R., Rashid, K.J., Rahman, S., Mia, M.J., Kanak, F.A., **Chowdhury, S.I.A.** and Jahan, M.N., 2023. *Effects of rising seas and geomorphological changes in the islands of northern Bay of Bengal. Regional Studies in Marine Science*, 68, p.103250. Elsevier.

Islam, S., Ferdous, L., **Chowdhury, S.I.A.** and Zaman, A.U., 2017. *Public and private hospital service quality assessment on the basis of patients' perception in Bangladesh: A case study in Rajshahi City Corporation (RCC). International Journal of Business, Social and Scientific Research (IJBSSR)*, 19, p.11.

Additional Skills

UAV and LiDAR: UAV flight planning & surveys; basic LiDAR handling.

Programming: Python (Arcpy, ArcGIS API for Python, Pytorch, Scikit-learn, GDAL, Geopandas, Pandas, Numpy, Pillow and so on), Javascript (ArcGIS Maps SDK for JavaScript, Leaflet, JQuery), SQL.

Data & Tools: ArcGIS Pro, ArcGIS Dashboards, ArcGIS Experience Builder, ArcGIS Survey 123, ArcGIS Field Maps, ArcGIS Drone2Map, ArcGIS StoryMaps, ArcGIS Urban, ArcGIS CityEngine, Qgis, MS Office, Autocad Map 3D, Google Sketchup, Adobe Illustrator, SPSS, Lumion 3D, Google Earth, GEE, Pix4d, Agisoft Metashape.

Languages

Bangla: Native command of spoken and written forms (C2).

English: Advanced professional command; competent in technical writing, presentations and cross-cultural communication (C1).

Trainings & Certifications

Esri (Instructor-Led/Web/Mooc): Creating Python Scripts for ArcGIS; Imagery Analysis in ArcGIS Pro; ArcGIS Pro Essential Workflows; Building Geodatabases; Deploying & Maintaining a Multiuser Geodatabase; Sharing GIS Content; Field Data Collection & Management; Spatial Data Science (MOOC); Python for Everyone; Raster Analysis Scripting; ArcGIS API for Python (Intro/Accessing Data/Performing Analysis); ArcGIS Maps SDK for JavaScript (Getting Started/Working with Maps); Deep Learning Using ArcGIS Pro; Classifying Objects Using DL in ArcGIS Pro; ArcGIS Drone2Map (Basics/Processing); Imagery & Remote Sensing (Getting Started); Managing/Displaying Raster Data.

Selected Training and Certifications: Operating UAV & USV (Airon Korea); Web GIS Programming Series (Udemy); GEE for Air Quality (Online).

Professional Membership

Bangladesh Institute of Planners (BIP): Member

The national professional organization of urban planners in Bangladesh, established in 1974 to promote sustainable planning practices, policy research, and professional development in urban planning.

Academic Honors

University Merit Scholarship (Undergrad).